

Do Foreign Mergers and Acquisitions Increase Acquired Firms' Innovation? Evidence from Chinese Manufacturing Firms

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Abstract

By merging China's manufacturing enterprise, patent and customs databases, we construct a sample of 27,130 firms from 2002–2007. Using Poisson fixed-effects regressions to control for firm and year heterogeneities and a host of control variables, we find that acquired firms of foreign mergers and acquisitions (FMA) apply for significantly more patents than their domestic counterparts in invention, utility model and design patents. Furthermore, FMA increases innovation through expanding firm size, exports and human capital formation. Wholly foreign-owned firms are more innovative than partially foreign-owned firms, and acquired firms in labour-intensive industries are more innovative than those in capital-intensive industries.

1. Introduction

A key reform policy that has been driving China's spectacular growth performance over the past 30 years is the encouragement of foreign direct investment (FDI) (Chen, Chang and Zhang 1995; Pomfret 1997; Liu and Wang 2003; Shan 2002; Yao 2006). In the early reform period, free trade zones were established in the coastal cities to attract foreign capital and absorb foreign technology. Since 1990, FDI has increased rapidly (40-fold), reaching US\$134 billion in 2016, the second highest in the world behind the United States (UNCTAD 2017). Given this large FDI inflow, it is worthwhile investigating whether FDI has been successful in enhancing China's innovation capability. After all, technological progress, as demonstrated in the growth literature, plays a crucial role in sustaining economic growth over the long term. In this article, we focus on a dominant type of FDI in developing countries: foreign mergers and acquisitions (FMA), and examine the empirical relationship between FMA and the innovation output of acquired firms.¹ In particular, this article attempts to address the following questions. First, are FMA an important vehicle for Chinese acquired firms to acquire the technological capability to innovate? Second, what are the mechanisms through which FMA help promote innovation for acquired firms? Third, what is the optimal mix of foreign–domestic capital structure for enhancing the innovation of acquired firms, wholly or partially foreign-owned firms?

The theoretical literature is ambiguous regarding how FMA affect the innovation of acquired firms. When one considers FMA and R&D activities, there are reasons to expect that FMA may reduce the R&D activities of

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acquired firms through rationalisation and elimination of duplicate R&D activities. Most firms conduct their R&D activities near their headquarters or corporate production units (UNCTAD 2005). In addition, FMA can lead to corporate restructuring and organisational changes with stricter financial controls, which may reduce R&D intensity (Hitt et al. 1991). By contrast, if FMA involve complementary technologies shared by both the acquiring and acquired firms, then the incentives to perform R&D may increase after an acquisition (Cassiman et al. 2005).

When one considers FMA and innovation output, there are reasons to expect that FMA enhance the innovation output of acquired firms. Markusen (1995) argues that the reason foreign parents transfer part of their technology to their affiliates is to exploit the affiliates' firm-specific assets in the host country. It is also believed that FMA increase the efficiency of R&D activities exploiting complementarities (Röller et al. 2001). On the contrary, Nocke and Yeaple (2008, p. 27), in their theoretical analysis, conclude that 'While FDI may positively affect the distribution of firm efficiencies in one country, it may have the opposite effect in the other country. Indeed, if productivity is measured at the plant level, cross-border acquisitions tend to reduce the observed productivity of the plants in the high cost country'. Thus, it is not clear from the theoretical standpoint how FMA affect the innovation output of acquired firms.

Previous studies in the area often find that FMA exert a positive impact on acquired firms' employment, total factor productivity or R&D intensity. As will be discussed in the following section, empirical studies that use micro firm-level data and direct measures of innovation output are sparse. Many previous studies draw conclusions based on differences in total factor productivity between FMA-acquired and non-acquired firms (see, for example, Doms and Jensen 1998; Griffith 1999). Since total factor productivity is calculated as a residual term, interpreting it as an indirect measure of innovation output is problematic, and it is also unclear whether a causal relationship can be established between FMA and total factor

productivity (Stiebale and Reize 2011). Other previous studies use R&D activities as a measure of innovation to find that FMA tend to increase the R&D activities of domestic Chinese firms (Liu and Zou 2008).² However, foreign equity participation in Chinese industrial firms can also weaken their own R&D as it becomes a substitute for absorbing market-mediated technology (Hu et al. 2005). Another concern is that R&D expenditure is highly correlated with firms' output level and may not reflect firms' incremental innovation (Hall et al. 2010). Thus, the empirical question of how FMA affect acquired firms' innovation remains unanswered.

This article contributes to the existing literature in several ways. First, in contrast to many previous studies, it uses micro firm-level data to discern the potential impact of FMA on the innovation output of acquired firms as measured by the number of patent applications. We are able to observe the change in ownership of any firms acquired by FMA and their performances in innovation, production, employment, exports, worker training and other activities over the sample period of 2002–2007. By using fixed-effects panel estimations to control for firm and time heterogeneities, we reduce the endogeneity concern that plagues the results of many previous studies. In particular, by using firm fixed-effects estimations, we can reduce the concern that firms that are more innovative are more likely to be acquired by multinational corporations. Second, this article sheds light on the mediating channels through which FMA affect the innovation of acquired firms. Some researchers have shown that firm size, exports and the extent of the firm's networking are important drivers of small and medium firms' innovation (Bloch and Bhattacharya 2016). This paper investigates whether these drivers of innovation are the mediating channels used by FMA to enhance the innovation of acquired firms in a developing country such as China. Third, we aim to shed light on the issue of whether wholly foreign-owned as opposed to partially foreign-owned firms are more conducive to innovation. It has been argued that 100 per cent foreign-owned investments are a more desirable type of FDI than joint ventures

for the host countries.³ It is our objective to provide some evidence from a large developing country on this issue.

The remainder of this article is organised as follows. Section 2 discusses previous studies in the literature to set the scene for the current study. Section 3 discusses the empirical approach in estimating the effects of FMA on firm innovation through the channels of firm size, exports and human capital formation. Section 4 defines the variables used in the regressions and discusses the construction of the dataset. Section 5 presents and discusses the regression results. Section 6 conducts further robustness checks, and Section 7 concludes the article.

2. Literature Review

2.1 FMA and Firm Innovation

Many previous studies have examined the effects of foreign ownership on acquired firms' innovation. In this literature, some research has shown that foreign ownership increases acquired firms' innovation activities (Lall 1983; Aw 2006; Ramstetter and Yang 2009). Arnold and Javorcik (2005), for example, use a dataset of Indonesian manufacturing firms to empirically analyse how FMA influence firm performance. Their results show that firms invested more in new capital equipment after being acquired by foreign firms. Guadalupe et al. (2012) construct a heterogeneous firm model to explain how foreign ownership affects innovation. Using a dataset of Spanish manufacturing firms between 1990 and 2006, they find that domestic firms that have been acquired by multinational corporations are more inclined to innovate. However, another stream of literature draws a different conclusion: foreign ownership has a negligible impact on domestic innovation (Rogers 2002; Lundin et al. 2007). For example, Stiebale and Reize (2011) use a large sample of small and medium-sized German firms to find that innovation output, measured as product and process innovations, and the share of sales from product innovations, is not significantly affected by a foreign acquisition for a given

amount of innovation efforts. They conclude that 'the estimation results do not provide any evidence of significant technology transfer through foreign acquisitions in the form of a higher innovation success' (p. 165).

2.2 Main Mechanisms of FMA on Innovation

2.2.1 FMA, Firm Size and Innovation

Some previous studies examine how FMA affect the size of acquired firms. For example, the bargaining power of labour unions in acquired firms is likely to be reduced as a result of the potential risk of divestment by the foreign parent firms. Falling wages would incentivise acquired firms to employ more workers and thus expand firm size (Brännlund et al. 2004). In addition, some local firms can better absorb the technology transferred by foreign capital mergers and acquisitions after foreign capital acquisitions, thereby stimulating the demand for labour and expanding the size of the company (Bandick and Görg 2010; Lipsey et al. 2010). Some studies have also found that expansion after acquisition is closely related to the host country's comparative advantage (see, among others, Conyon et al. 2010; Girma et al. 2013). Moreover, a change in firm size can also be linked to both horizontal and vertical FMA. In horizontal FMA, foreign firms capture the host country's market by transferring technology and capital from the parent firm to the acquired firm thereby increasing the latter's employment and output. In vertical FMA, on the other hand, the motivation of foreign firms to integrate the acquired firms into their production chain is likely to improve the efficiency of acquired firms and increase their size (Bobonis and Shatz 2007; Ajaga and Nunnenkamp 2008).

Results of previous studies are inconclusive on the issue of whether larger firms are more innovative. Schumpeter (1942) postulates that larger firms have an advantage over smaller firms in engaging innovation because they have the resources and incentives to do so. However, Scherer (1965, 1967) finds that the relationship between firm size and innovation is not linear, but rather has an inverted U shape. More

recently, Shefer and Frankel (2005) suggest that firm size has a negative effect on R&D expenditure in high-tech industries, but that this effect is insignificant in traditional industries. Bhattacharya and Bloch (2004) find a quadratic relationship between firm employment and firm innovation as measured by a dichotomous variable, namely, whether or not the firm has introduced an innovation in products (including services for the non-manufacturing firms) or processes. Cohen (2010) summarises the literature on the relationship between firm size and innovation as follows: 1) the likelihood of conducting R&D increases with firm size; 2) R&D rises monotonically with firm size (Baldwin and Scott 1987); and 3) innovation per unit of R&D tends to decline with firm size (Scherer 1965; Pavitt 1987; Acs and Audretsch 1988, 1990). Thus, larger firm size may not necessarily increase innovation, although it tends to increase R&D.

2.2.2 *FMA, Exports and Innovation*

FMA are likely to increase exports by acquired firms. Hanson et al. (2005) indicate that the United States' foreign affiliates import intermediate inputs from their parent firms and then export final products to the United States. Such vertical production networks form the parent-to-affiliate trade model in the United States. Ekholm et al. (2007) investigate the behaviour through which affiliates export to third-party countries rather than to the parent or host countries. They find that the horizontal production model between affiliates and their parent firms dominates export markets and that vertical production complements it. Guadalupe et al. (2012) describe the above situations as multinational corporations providing export markets to their acquired overseas subsidiaries (including vertical and horizontal FMA). Their empirical results show that the share of exports in sales revenue increases significantly in the acquired firms. FMA effectively enhance the export strength of the acquired firms. Lipsey et al. (2010) find evidence that foreign-owned firms are better able to penetrate export markets due to their larger production scale and greater efficiency. They conclude that FMA increase

the level of acquired firms' exports. Using data from Chinese manufacturing firms between 1999 and 2005, Du and Girma (2009) show that FMA only enhance exports if the acquired firm is less than 45 per cent foreign-owned. When foreign ownership is higher than 45 per cent, exports are reduced.

Regarding the link between exports and innovation, much research has already been carried out to show that developing countries can gain technology and knowledge spill-overs by trading with developed countries (Shefer and Frankel 2005; Atkeson and Burstein 2010). There are three main reasons for this. First, trade expansion reduces international transport and operating costs, which in turn stimulate R&D and innovation efforts (Grossman and Helpman 1991; Baldwin and Gu 2004). Second, the export sector can establish forward and backward linkages with the domestic sector, accelerating technology diffusion and motivating independent innovation (Impullitti and Licandro 2017). Third, fierce market competition forces exporting firms to continuously innovate to increase their competitive ability (Canto and González 1999; Li et al. 2008). Meanwhile, the shift of production resources to firms with high productivity can also improve these firms' innovation levels (Aw and Hwang 1995; Foster et al. 2008). However, some studies show that exporting firms in developing countries could be stuck in low-tech production because these exporting firms are located at the low end of the global value chain, which makes high-end upgrading and innovation difficult (Schmitz 2004; Zhang et al. 2009). In addition, a developing country is required to achieve a certain minimum threshold in income level, human capital formation and institutional quality before exporting firms in the country can engage in innovation (Moschos 1989; Aw et al. 2011).

2.2.3 *FMA, Human Capital Formation and Innovation*

Finally, FMA can enhance the human capital formation of workers in acquired firms. Using 1980s' data from Portugal, Mata and Portugal (2004) find that newly established

foreign-owned firms or FMA-acquired firms had a more educated workforce compared to domestic firms. Conyon et al. (2010) show that the average wage level increased by 3.4 per cent when firms were acquired by FMA. A study using Indonesian data by Lipsey and Sjöholm (2002) finds that wages for blue and white collared workers increased by 10 per cent and 21 per cent respectively after their firms were acquired by FMA. Guadalupe et al. (2012) also find that acquired firms experience an improvement in the skills level of their workforce. However, using Portuguese data between 1991 and 1998, Almeida (2007) finds that FMA do not appear to generate any significant improvement in education or increases in wages for workers in the acquired firms.

Regarding the link between human capital formation and innovation, endogenous growth theory emphasises the positive role played by human capital in innovation (Grossman and Helpman 1991). Both education and on-the-job training contribute to human capital formation. Education can strengthen employees' ability to accept and understand new knowledge, thus promoting the adoption of new technologies (Nelson and Phelps 1966; Benhabib and Spiegel 1994). Dostie (2013) uses detailed data on firms' human capital investments and innovation performance, and finds that more training leads to more product and process innovation, with on-the-job training playing a role that is as important as classroom training. González et al. (2016) confirm that R&D is more effective when firms also invest in worker training. Moreover, for some smaller firms, investing in workers' skills (via training programs) boosts innovation even in the absence of R&D. Studies have also shown that on-the-job training improves innovation (Baldwin and Johnson 1996; Ballot et al. 2001; Bauernschuster et al. 2009; Caloghirou et al. 2018). Hashimoto (1991) also finds that the effective use of new technology is highly correlated with employee training in Japan.

2.3 Innovation and Patents

In the innovation process, patents are usually treated as an output of the knowledge

production function, and an input to the production function to explain a firm's behaviour, such as productivity (Griliches 1990). The number of patents is a proxy for knowledge capital, which can be used as one of the input factors in a firm's production function, and can also be used as an innovation indicator (Pakes and Griliches 1984). However, not all inventions are patented (Levin et al. 1987; Cohen et al. 2002), and not all patents are used as an input to the production of firms (Gambardella et al. 2007; Motohashi 2008). Also, patent data have some limitations as an indicator of innovation (Nagaoka et al. 2010).

In the literature, investment in R&D, the number of R&D personnel and new product sales are the most widely used measures of firm innovation (Jefferson et al. 2006; Leenders and Wierenga 2008; Jankowski 2012). However, investment in R&D and the number of R&D personnel are highly correlated with firms' output levels and thus cannot accurately reflect the nature of incremental development (Hall et al. 2010). Sales of new products embody the commercialisation of R&D output, and thus are also affected by firms' output levels. In contrast, the patent data are relatively objective, and can more accurately reflect the current level of firms' innovation (Hall and Harhoff 2012; Li et al. 2016).

In conclusion, we note that there are existing studies that link FMA directly to the innovation of acquired firms, FMA to firm size, FMA to exports and FMA to human capital formation. Many studies have also been carried out on linking firm size, exports and human capital formation on the one hand to firm innovation on the other. However, the literature on how firm size, exports and human capital formation act as mediating channels for FMA to affect the innovation of acquired firms is still sparse, and this will be the main focus of our attention in the remaining sections of the article.

3. Method

We postulate the following empirical model based on Guadalupe et al. (2012):

$$p_{i,t} = \beta_0 + \beta_1 b_{i,t-1} + \beta_2 X_{i,t} + \varepsilon_{i,t} \quad (1)$$

where i and t refer to firm i in year t , and p is the dependent variable for the total number of patent applications. The explanatory variable $jb_{i,t-1}$ denotes one-year lagged FMA. X is a vector of control variables affecting patent applications, and ε is the error term.

To investigate the mediating channels through which FMA affect firm innovation as described in Section 2, we first check whether firm size, exports and human capital formation are linked to FMA as follows:

$$lnem_{i,t} = \alpha_0 + \alpha_1 jb_{i,t-1} + \alpha_2 X_{i,t} + \omega_{i,t} \quad (2)$$

$$lnex_{i,t} = \gamma_0 + \gamma_1 jb_{i,t-1} + \gamma_2 X_{i,t} + \tau_{i,t} \quad (3)$$

$$lnwt_{i,t} = \theta_0 + \theta_1 jb_{i,t-1} + \theta_2 X_{i,t} + \epsilon_{i,t} \quad (4)$$

where i and t refer to firm i in year t , and the left-hand side variables $lnem$, $lnex$ and $lnwt$ are respectively firm size, exports and human capital formation (measured by investment in on-the-job-training). The explanatory variables $jb_{i,t-1}$ denote one-year lagged FMA, and X is a vector of control variables. Finally, ω , τ and ϵ are the error terms.

Our attempt to show how FMA mechanisms affect firm innovation requires that FMA have a significant positive effect on firm innovation in Equation (1), but this positive effect of FMA on firm innovation is expected to be substantially reduced by the presence of the mediating variables of firm size, exports and human capital formation in the following regression:

$$p_{i,t} = \pi_0 + \pi_1 jb_{i,t-1} + \pi_2 lnem_{i,t} + \pi_3 lnex_{i,t} + \pi_4 lnwt_{i,t} + \mu_{i,t} \quad (5)$$

Equations (1) to (5) encompass an empirical approach for testing the presence of mediators as discussed by Baron and Kenny (1986). If firm size, exports and human capital formation are the mediators between FMA and firm innovation, it is expected that: 1) the estimated coefficient of FMA, $jb_{i,t-1}$, should be significant in Equations (1) to (4); 2) all the estimated coefficients of the mediators ($lnem$, $lnex$ and $lnwt$) should be significant in Equation (5); and 3) the coefficient of FMA,

$jb_{i,t-1}$, in Equation (5) should be less than that in Equation (1). When the above three conditions are satisfied, the mediating effects of firm size, exports and human capital formation can then be established. Furthermore, if the estimated coefficient of FMA in Equation (5) remains statistically significant, firm size, exports and human capital formation do not account for all of the effects of FMA on firm innovation, indicating that FMA have some other additional direct or indirect effects on firm innovation. If, however, the estimated coefficient of FMA in Equation (5) becomes statistically insignificant, then we can conclude that firm size, exports and human capital formation are the mediators that account for all of the effects of FMA on firm innovation.

4. Data

4.1 Measure of Firm Innovation

The firm-level patent dataset used in this article comes from the ‘China Patent Database Digest 1985–2012’ published by the State Intellectual Property Office (SIPO) Safety Publishing House. The database includes approximately 7.7 million patent applications in three categories of patents since the implementation of the Patent Law in September 1985. There are three different types of patent applications: invention, utility model and design. The definition of invention patents is ‘any new technical solution relating to a product, a process or improvement’ (SIPO, 2016, pp. 23–25). Utility model patents are defined as ‘any new technical solution relating to the shape, the structure, or their combination, of a product, which is fit for practical use’ (SIPO 2016, pp. 23–25). Design patents are ‘any new design of the shape, the pattern or their combination, or the combination of the colour with shape or pattern, of a product, which creates an aesthetic feeling and is fit for industrial application’ (SIPO 2016, pp. 23–25). Invention patent applications are the most technologically demanding of the three types of patent applications. We use the number of patent applications as an innovation indicator.

4.2 Measure of FMA and Firm Characteristics

The data on firm characteristics are obtained from the Chinese Industrial Enterprises 2002–2007 database. Our sample excludes all firms that show a negative or zero value for total revenue, employment, fixed assets and total sales. Also, firms with less than eight workers are excluded. Moreover, firms whose total sales are less than export sales, current assets are greater than total assets, and level and value of fixed assets are greater than total assets are excluded from the sample.

In China, a firm is officially defined as foreign-owned if foreign capital accounts for more than 25 per cent of the firm's total capitalisation. The FMA variable, jb , is thus a dummy variable which takes the value of one if foreign capital accounts for more than 25 per cent of the firm's total capitalisation, and zero otherwise. Foreign capital includes capital from abroad or from Hong Kong, Macao and Taiwan. To identify the impact of foreign ownership in the analysis, firms that have already been defined as foreign-owned in 2000 are excluded from the sample. That is, those firms with a foreign ownership share above 25 per cent of their total capitalisation in 2000 are excluded. This strategy enables us to observe a switch of ownership from domestic to foreign for those acquired firms during the sample period of 2001–2006. Also, it is unlikely that FMA can exert their potential effects on firm innovation immediately after a takeover, and thus a one-year lagged FMA, $jb_{i,t-1}$, is used in the regression. Moreover, using the lagged FMA rather than the current FMA has the advantage of reducing endogeneity between FMA and firm innovation (Guadalupe et al. 2012).

The size of a firm is measured by its employment, em , in logarithms, $Ln(em+1)$. Exports, ex , is measured by export delivery values. As export data are missing for 2004 from the Chinese Industrial Enterprises database, we obtain the missing export data by merging the Chinese Industrial Enterprises database with the Chinese Customs database. Owing to the unavailability of data on workers'

educational attainment, human capital formation, wt , is measured by a firm's investment in on-the-job training. Investment in on-the-job training is deflated by China's provincial PPI, and measured in logarithms, $Ln(wt+1)$.

In addition to the above explanatory variables, we also add firm productivity (TFP), capital intensity (K/L), government subsidies (sub), firm age (age), profit level (pro) and liability-to-capital ratio ($Liability/Asset$) in the regressions as controls. Firm productivity, ϕ , is measured by the firm's total factor productivity (TFP) growth. Following Brandt et al. (2012), TFP growth is calculated by subtracting the growth of all inputs from the growth of total value added. Value added and intermediate inputs are deflated by China's provincial PPI with 2000 as the base year. Furthermore, in order to reduce the concern of reverse causality between patent applications and TFP growth, TFP growth is lagged by one year in the regressions. The capital intensity of a firm is measured by its capital-to-labour ratio. Capital-intensive firms are likely to benefit more from innovation than less capital-intensive firms, and thus they may spend more resources on R&D and produce more patent applications.

Since undertaking innovation activities is risky for firms, governments may use industrial policies, fiscal policies and other means to encourage and promote firm innovation. It is worth noting that government subsidies may have complementary or substitutionary effects on firm innovation. On the one hand, government subsidies can address the issues of knowledge leakage and incomplete exclusivity in R&D activities (Stiglitz 1987). On the other hand, government subsidies may generate crowding-in effects on R&D investment and inhibit firm innovation.

Firm age is the number of years since the establishment of the firm. There are two opposing effects of firm age on innovation. On the one hand, the older the firm, the more experience it has in its own innovative activities, which can lower the cost of innovation. On the other hand, older firms tend to maintain their existing range of products due to their sunk cost and therefore may have a lower propensity to engage in innovation. Firm profit

level is measured by the ratio of profit to the total output value. R&D activities can incur high costs, and profitable firms are in a better financial position to invest in R&D. The liability-to-capital ratio is measured by the ratio of total liabilities to total assets. Existing theory suggests that highly indebted firms are less likely to take risk and to invest in R&D (Wang 2002). Nevertheless, new start-up firms may incur high debt in order to develop new products or processes. Summary statistics and definitions of all the variables used in the regressions are presented in Table 1.

5. Results

5.1 Baseline Results

This section reports the regression results of Equation (1) for the total number of patent applications and for each of the three different types of patent applications (invention, utility model and design) as a dependent variable. The Hausman test rejects the panel random-effects estimator, and thus the Poisson fixed-effects panel estimator is used (Cameron and Trivedi 2005).⁴ We use both the firm and year fixed-effects in the regressions to control for all unobserved time-invariant firm heterogeneities and for any year-specific factors affecting all firms.

Column 1, Table 2, shows the results when the total number of patent applications is used as the dependent variable. First, holding all other variables constant, a change in a firm's ownership from domestic to foreign in the previous year is estimated on average to be associated with an increase in the total number of patent applications of this firm by approximately 0.548, which is statistically significant at the 1 per cent level. This result provides strong evidence to support that FMA promote firm innovation. Second, columns 2, 3 and 4 of Table 2 show that all three types of patent applications (invention, utility model and design patents) are significantly higher with a change in a firm's ownership from domestic to foreign in the previous year. More specifically, when the ownership of a firm switches from domestic to foreign in the previous year, applications for invention patents, which are most indicative of the technological and innovative capacity of a firm, are estimated to experience the largest increase (0.796 patents) compared to increases in applications for utility model patents (0.440 patents) and design patents (0.742 patents). One possible explanation is that utility model and design patents do not require as much technological knowhow as invention patents and thus domestic firms do not necessarily have to rely on technology transfer from FMA for

Table 1 Summary Statistics

Variable	Definition	Obs.	Mean	Std. dev.	Min.	Max.
p	Total no. of patent applications	162,780	0.320	18.295	0	4975
fm	Invention patent applications	162,780	0.131	16.881	0	4662
sy	Utility model patent applications	162,780	0.096	1.861	0	296
wg	Design patent applications	162,780	0.092	2.535	0	617
$Jb_{i,t-1}$	Lagged FMA, $Jb_{i,t-1} = 1$ if a firm is more than 25% foreign owned, 0 otherwise	162,780	0.028	0.166	0	1
$Ln(em)$	No. of workers in log	162,780	5.353	1.150	2.079	11.964
$Ln(ex+1)$	Export value+1 in log	162,780	0.262	0.440	0	17.871
$Ln(wt+1)$	Per worker on-the-job training investment+1 in log	162,780	0.113	0.186	-1.727	5.632
$Ln(tfpi,t-1)$	Lagged total factor productivity in log	162,780	6.823	1.140	-7.282	13.623
$Ln(k+1)$	Capital per worker+1 in log	162,780	3.861	1.236	-11.933	13.104
$Ln(sub+1)$	Per worker government subsidy+1 in log	162,780	0.266	0.710	-1.901	6.535
$Ln(age+1)$	Firm age+1 in log	162,780	2.682	0.746	0	6.011
pro	Ratio of profit to total value of output	162,780	0.058	1.296	1.99e-4	75.146
$Liability/Asset$	Ratio of total liabilities to total assets	162,780	0.607	0.282	2.11e-4	15.116

Notes: See Section 4 for sources and descriptions of data in detail.

Table 2 Effect of FMA on Innovation: Baseline Results

	<i>Total patents (p)</i>	<i>Invention (fm)</i>	<i>Utility model (sy)</i>	<i>Design (wg)</i>
	(1)	(2)	(3)	(4)
FMA	0.548***	0.796***	0.440**	0.742***
Jb _{i,t-1}	(0.2061)	(0.2735)	(0.2177)	(0.2250)
TFP	0.304***	0.300***	0.215***	0.212***
Ln(tfpi,t-1)	(0.0422)	(0.0620)	(0.0485)	(0.0581)
K/L	0.153***	0.274***	0.080*	0.001
lnk+1	(0.0543)	(0.0669)	(0.0473)	(0.0596)
Subsidy	-0.051	-0.014	0.074**	0.049
Ln(sub+1)	(0.0467)	(0.0482)	(0.0375)	(0.0597)
Age	-0.135**	-0.054	-0.077	-0.172**
Ln(age+1)	(0.0595)	(0.1160)	(0.0852)	(0.7223)
Profit	-0.414	-0.557	-0.119	-0.023
(pro)	(0.3191)	(0.5488)	(0.1963)	(0.0326)
Liability/	0.200	0.694**	-0.110	-0.230
Asset	(0.1703)	(0.3045)	(0.2120)	(0.2658)
Firm FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Obs.	162,780	162,780	162,780	162,780

Notes: ***, ** and * indicate significance at the 1 per cent, 5 per cent and 10 per cent levels, respectively. Standard errors are in parentheses. The dependent variable is the total number of patent applications in column 1, invention patent applications in column 2, utility model patent applications in column 3 and design patent applications in column 4. Regressions use the Poisson fixed-effects panel estimator.

engaging in such innovative activities. Consequently, FMA play a relatively minor role in utility model and design patent applications.

Table 2 also shows that firm productivity is positively associated with the total number of patent applications and with each of the three types of patent applications. These results provide evidence to support that more productive firms tend to apply for more patent applications. Capital intensity is positively associated with the total number of patent applications, invention and utility model patents, but has little association with design patents. Government subsidy is only positively associated with utility model patents. Firm age is estimated to have a negative association with the total number of patent applications and design patent applications which are statistically significant at the 5 per cent level. This suggests that older firms tend not to engage in innovative activities that change the shape and/or structure of their existing products. Profit level appears to have little association with patent applications. The liability–asset ratio is only positively associated with invention patents.

5.2 FMA, Firm Size, Exports and Human Capital Formation

Table 3 reports the estimation results of Equations (2) to (4). Column 1, Table 3, shows the results when firm size is used as the dependent variable. Fixed-effects panel estimation results using robust standard errors and clustering at the firm level show that the estimated coefficient of FMA is 0.068, which is statistically significant at the 1 per cent level, indicating that the size of foreign-acquired firms expanded after a FMA takeover. Holding all other variables constant, a change in a firm's ownership from domestic to foreign in the previous year is estimated on average to be associated with an increase in the total number of employees of this firm by approximately 0.070 ($e^{0.068}-1$) people.

Columns 2 and 3, Table 3, show the results when exports and investment in on-the-job training are the dependent variables. The estimated coefficients of FMA are respectively 0.242 and 0.002 for columns 2 and 3, both of which are statistically significant at the conventional levels, indicating that exports

Table 3 Firm Size, Exports, Human Capital Formation and FMA

	Size (<i>lnem</i>)	Exports (<i>lnex+1</i>)	Human capital (<i>lnwt+1</i>)
	(1)	(2)	(3)
FMA	0.068***	0.242***	0.002*
$Jb_{i,t-1}$	(0.0115)	(0.0740)	(0.0011)
TFP	0.083***	0.117***	0.010***
$\ln(tfpi,t-1)$	(0.0023)	(0.0101)	(0.0009)
K/L	-0.210***	-0.016	0.017***
$\ln k+1$	(0.0065)	(0.0123)	(0.0013)
Subsidy	-0.021***	-0.007	0.008***
$\ln(sub+1)$	(0.0028)	(0.0129)	(0.0014)
Age	0.033***	-0.010	-0.0001
$\ln(age+1)$	(0.0334)	(0.0185)	(0.0012)
Profit	-0.001	-0.0003	0.0005
(pro)	(0.0018)	(0.0005)	(0.0006)
Liability/	-0.031**	0.081**	0.003
Asset	(0.0097)	(0.0369)	(0.0028)
Firm FE	yes	yes	yes
Year FE	yes	yes	yes
Obs.	162,780	162,780	162,780

Notes: ***, ** and * indicate significance at the 1 per cent, 5 per cent and 10 per cent levels, respectively. Heteroscedastic robust standard errors are in parentheses. The dependent variable is firm size in column 1, exports in column 2 and human capital in column 3. Regressions use OLS fixed-effects panel estimator.

and investment in on-job-training of foreign-acquired firms increased after a FMA takeover. Specifically, holding all other variables constant, a change in a firm's ownership from domestic to foreign in the previous year is estimated on average to be associated with increases in the firm's exports and investment in on-job-training by approximately ¥2.74 thousand and ¥0.02 thousand respectively. In summary, the results in Table 3 show that firm size, exports and human capital formation increased significantly after a FMA takeover.

5.3 Mechanisms of FMA on Firm Innovation

Table 4 reports the estimation results of Equation (5). Column 1, Table 4, shows the results when the total number of patent applications is used as the dependent variable. The three mediating variables *lnem*, *lnex* and *lnwt* are all significantly positive, and the estimated coefficient of FMA, $jb_{i,t-1}$, is also statistically significant, but is less than that in Table 2 (0.548 versus 0.526). As discussed in Section 3, these results suggest a partial mediating effect of firm size, exports and

human capital formation. That is, FMA promote firm innovation by increasing firm size, exports and human capital formation, but they also affect firm innovation either directly or through some other channels.

Column 2, Table 4, shows that firm size and exports are significantly correlated with invention patent applications while human capital formation is not, indicating that firm size and exports are mediators of FMA for invention patents, but not human capital formation. Column 3, Table 4, shows that only firm size is significantly positive. Thus, only firm size can be established as a mediator of FMA for design patents. In column 4, the estimated results of design patents are similar to those of the total number of patent applications, indicating that the same three mechanisms of FMA for design patents can also be established.

5.4 Wholly Foreign-Owned versus Partially Foreign-Owned Firms

An interesting question to be addressed is that if FMA stimulate acquired firms' innovation, would a wholly foreign-owned firm such as a

Table 4 Testing the Mediating Effects of Firm Size, Exports and Human Capital Formation

	<i>Total patents (p)</i>	<i>Invention (fm)</i>	<i>Utility model (sy)</i>	<i>Design (wg)</i>
	(1)	(2)	(3)	(4)
FMA	0.526***	0.782***	0.434**	0.633***
Jb _{i,t-1}	(0.2024)	(0.2733)	(0.2195)	(0.2166)
Size	0.612***	0.261**	0.456***	0.767***
Ln(em+1)	(0.0925)	(0.1069)	(0.1063)	(0.1216)
Exports	0.030***	0.036***	0.007	0.028**
Ln(ex+1)	(0.0085)	(0.0139)	(0.0093)	(0.0134)
Human Cap.	0.014*	-0.142	0.062	0.660***
Ln(wt+1)	(0.0076)	(0.1425)	(0.1037)	(0.2325)
TFP	0.214***	0.235***	0.155***	0.175***
Ln(tfpi,t-1)	(0.0365)	(0.0560)	(0.0471)	(0.0510)
K/L	0.055	0.116*	0.204***	0.153**
lnk+1	(0.0533)	(0.0624)	(0.0590)	(0.0698)
Subsidy	0.032	0.003	0.074**	0.056
Ln(sub+1)	(0.0377)	(0.0494)	(0.0375)	(0.0545)
Age	-0.151***	-0.071	-0.104	-0.159**
Ln(age+1)	(0.0595)	(0.1053)	(0.0848)	(0.0729)
Profit	-0.235	-0.341	-0.142	0.006
(pro)	(0.2590)	(0.3747)	(0.1841)	(0.0366)
Liability/	0.123	0.440	-0.137	0.004
Asset	(0.1744)	(0.2742)	(0.1924)	(0.2588)
Firm FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Obs.	162,780	162,780	162,780	162,780

Notes: ***, ** and * indicate significance at the 1 per cent, 5 per cent and 10 per cent levels, respectively. Standard errors are in parentheses. The dependent variable is the total number of patent applications in column 1, invention patent applications in column 2, utility model patent applications in column 3 and design patent applications in column 4. Regressions use the Poisson fixed-effects panel estimator.

greenfield investment made by a foreign firm engage in even more innovation than a FMA-acquired firm? It is reasonable to assume that multinational corporations are more inclined to conduct R&D activities in wholly foreign-owned firms because they can better protect the results of their innovation against leakage to other firms than if they only own a fraction of the acquired firms in the host country. Yet, wholly foreign-owned firms may have less incentive to apply for patent rights given that applying for patents requires the revelation of the details of innovation to the public in the host country. We aim to shed light on this question by dividing our dataset into two subsamples. The first subsample includes only those firms that became wholly foreign-owned and those firms that remained in domestic ownership during the sample period. The second subsample includes those firms that became partially foreign-owned and those

firms that remained in domestic ownership during the sample period. A partially foreign-owned firm is defined as one in which foreign capital accounts for more than 25 per cent but less than 100 per cent of the firm's total capitalisation. Regression results of this exercise are presented in Table 5.

Table 5 shows clearly that wholly foreign-owned firms outperform partially foreign-owned firms in the total number of patent applications and in each of the three different types of patent applications. In column 2, it is estimated that switching ownership from domestic to partially foreign-owned increases the total number of patent applications by 0.427 patents, which is statistically significant at the 10 percent level, whereas in column 1, switching ownership from domestic to wholly foreign-owned results in an increase of 1.199 patents, which is statistically significant at the 1 per cent level. Similarly, invention, utility

Table 5 Results of Wholly versus Partially Foreign-Owned FMA

	Total patents (p)		Invention (fm)		Utility model (sy)		Design (wg)	
	Wholly	Partially	Wholly	Partially	Wholly	Partially	Wholly	Partially
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
FMA	1.199***	0.427*	1.457***	0.773**	1.319***	0.344	1.277***	0.577**
Jb _{i,t-1}	(0.2191)	(0.2248)	(0.3922)	(0.3089)	(0.2508)	(0.2362)	(0.3877)	(0.2457)
Controls	yes	yes	yes	yes	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes	yes	yes	yes
Obs.	159,691	161,262	159,691	161,262	159,691	161,262	159,691	161,262

Notes: ***, ** and * indicate significance at the 1 per cent, 5 per cent and 10 per cent levels, respectively. Standard errors are in parentheses. The sample of wholly foreign-owned firms includes 100% foreign-owned firms and domestic firms, but excludes partially foreign-owned firms. The sample of partially foreign-owned firms includes partially foreign-owned firms and domestic firms, but excludes 100% foreign-owned firms. The dependent variable is the total number of patent applications in columns 1 and 2, invention patent applications in columns 3 and 4, utility model patent applications in columns 5 and 6 and design patent applications in columns 7 and 8. Controls are lagged TFP, K/L, government subsidy, firm age, profit level and liability–asset ratio. To save space, we do not report the results of these control variables, but can provide them upon request. Regressions use the Poisson fixed-effects panel estimator.

model and design patent applications are all estimated to increase significantly when ownership is switched from domestic to wholly foreign-owned, and their coefficients are larger than those whose ownership is switched from domestic to partially foreign-owned. As discussed above, these results suggest that wholly foreign-owned subsidiaries are more inclined to conduct R&D activities and to apply for patents.

5.5 Technology-Intensive versus Capital-Intensive versus Labour-Intensive Firms

FMA-acquired firms in a particular industry may behave differently in their innovative activities from FMA-acquired firms in another industry. To address this issue, we divide the dataset into three subsamples according to whether the firm in question is in the technology-intensive, capital-intensive or labour-intensive sector. Following Wang (2004), we classify firms from 36 out of a total 40 two-digit industries into one of three industrial sectors.⁵

Table 6 reports the regression results which indicate that FMA-acquired firms in labour-intensive industries are estimated to have a significantly highest total number of patent applications and also utility and design patent

applications. For FMA-acquired firms in capital-intensive industries, they are estimated to have a significantly higher total number of patent applications and also invention and utility model patent applications, but not design patent applications. For FMA-acquired firms in technology-intensive industries, only design patent applications are estimated to be significantly higher.

In summary, the results in Table 6 show that acquired firms in labour-intensive industries are most affected by FMA, and acquired firms in technology-intensive industries are least affected by FMA. For acquired firms in capital-intensive industries, FMA are also an important impetus for firm innovation. A puzzling question is why are acquired firms in technology-intensive industries least affected by FMA? The reason is likely that multinational corporations in high-tech industries are subject to restrictions on technology exports imposed by their respective home countries (Europe, Japan and the United States). These restrictions protect against frontier ‘strategic’ technology and advanced production equipment to be transferred to China. Europe and the United States impose strict export restrictions on high-tech products to China, and only allow exports of technological products to China which are relatively backward (National Science Board

Table 6 Results of Technology-, Capital- and Labour-Intensive Firms

	<i>Total patents (p)</i>			<i>Invention (fm)</i>		
	<i>Technology</i>	<i>Capital</i>	<i>Labour</i>	<i>Technology</i>	<i>Capital</i>	<i>Labour</i>
	(1)	(2)	(3)	(4)	(5)	(6)
FMA	0.147	1.418***	1.527***	0.416	1.715***	1.700***
Jb _{i,t-1}	(0.2126)	(0.3537)	(0.3608)	(0.2825)	(0.3948)	(0.2756)
Controls	yes	yes	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes	yes
Observations	86,625	25,600	45,679	86,625	25,600	45,679
	<i>Utility Model (sy)</i>			<i>Design (wg)</i>		
	<i>Technology</i>	<i>Capital</i>	<i>Labour</i>	<i>Technology</i>	<i>Capital</i>	<i>Labour</i>
	(7)	(8)	(9)	(10)	(11)	(12)
FMA	0.093	1.892***	2.368***	0.353*	-0.216	1.153**
Jb _{i,t-1}	(0.2375)	(0.6064)	(0.3417)	(0.1941)	(0.4384)	(0.4913)
Controls	yes	yes	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes	yes
Observations	86,625	25,600	45,679	86,625	25,600	45,679

Notes: ***, ** and * indicate significance at the 1 per cent, 5 per cent and 10 per cent levels, respectively. Standard errors are in parentheses. Technology refers to the sample of firms in technology-intensive industries, Capital refers to the sample of firms in capital-intensive industries and Labour refers to the sample of firms in labour-intensive industries. See Section 4.5 for definitions of these three industrial groups. The dependent variable is the total number of patent applications in columns 1, 2 and 3, invention patent applications in columns 4, 5 and 6, utility model patent applications in columns 7, 8 and 9 and design patent applications in columns 10, 11 and 12. Controls are lagged TFP, K/L, government subsidy, firm age, profit level and liability–asset ratio. To save space, we do not report the results of these control variables, but can provide them upon request. Regressions use the Poisson fixed-effects panel estimator.

2016). For example, Science and Engineering Indicators 2016, prepared by the National Science Board (2016), show that domestic companies in China's pharmaceutical industry mainly produce non-patented drugs, while advanced production equipment is controlled by multinational corporations. Also, Japanese automobile manufacturers in China do not transfer frontier production technology to China's automotive industry (Nicholas et al. 2011). Therefore, even if firms in high-tech industries in China are acquired by foreign firms, there are still barriers to the acquired firms obtaining advanced production technologies and production equipment from Europe, Japan and the United States.

6. Further Robustness Checks

In this section, we conduct further analyses to check the robustness of our baseline results.

In the baseline results, we use one-year lagged FMA to allow for the impact of FMA on firm innovation to take effect. One concern is that one year may be too short a time frame to observe the potential impact of FMA on firm innovation. An additional concern is whether the baseline results are robust to an alternative definition of FMA since the baseline results are based on FMA being defined as foreign ownership accounting for 25 per cent or more of a domestic firm. We address both of these concerns in turn below.

Once a domestic firm is acquired by FMA, implementation of new technology and/or production methods requires training of workers that may involve substantial adjustments to existing work practices and culture. To address such a concern, we conduct a robustness check by lagging FMA by two years rather than one year in Equation (1).

Table 7 Using Two-Year Lagged FMA for Robustness Check

	<i>Total patents (p)</i>	<i>Invention (fm)</i>	<i>Utility model (sy)</i>	<i>Design (wg)</i>
	(1)	(2)	(3)	(4)
FMA	0.434*	0.760*	0.581**	0.800***
Jb _{i,t-2}	(0.2330)	(0.4522)	(0.2578)	(0.2368)
TFP	0.240***	0.132*	0.151**	0.246***
Ln(tfpi,t-1)	(0.0571)	(0.0787)	(0.0595)	(0.0681)
K/L	0.152**	0.250***	0.116*	0.007
lnk+1	(0.0603)	(0.0677)	(0.0603)	(0.0690)
Subsidy	0.119	0.207***	0.377***	0.117*
Ln(sub+1)	(0.0801)	(0.0725)	(0.0610)	(0.0626)
Age	-0.099*	0.067	0.037	-0.209**
Ln(age+1)	(0.0540)	(0.0856)	(0.0589)	(0.0830)
Profit	0.015	0.151	-0.032	0.364
(pro)	(0.2075)	(0.1991)	(0.1809)	(0.3249)
Liability/	1.158***	1.778***	0.927***	0.160
Asset	(0.2906)	(0.3647)	(0.3050)	(0.2901)
Firm FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Obs.	131,754	131,754	131,754	131,754

Notes: ***, ** and * indicate significance at the 1 per cent, 5 per cent and 10 per cent levels, respectively. Standard errors are in parentheses. The dependent variable is the total number of patent applications in column 1, invention patent applications in column 2, utility model patent applications in column 3 and design patent applications in column 4. Regressions use the Poisson fixed-effects panel estimator.

The results are reported in Table 7. Results show that lagging FMA by two years reduces the significance of the estimates of FMA on patent applications in comparison with those baseline results. However, the FMA estimates remain statistically significant for the total number of patent applications and for all three types of patent applications. Also, it is worth noting that by lagging FMA by two years, the number of firms included in the sample dropped by 5,171. It is thus reassuring that the baseline results hold to a large extent despite using a much smaller sample size and using a longer lag for FMA.

Another concern is whether the baseline results are robust to alternative definitions of FMA. According to the official Chinese definition, a firm is foreign-owned if foreign capital accounts for more than 25 per cent of the firm's capitalisation. However, the definition used by Guadalupe et al. (2012) is that a domestic firm is considered to be acquired by a foreigner if foreign capital accounts for more than 50 per cent of the firm's total capitalisation. We now check the robustness of our baseline results by using this alternative

definition of FMA, and the results are reported in Table 8. Results show that using the higher share of foreign capital as a definition of FMA does not substantially alter the baseline results. Despite the size of estimates being smaller for FMA, they remain positive and statistically significant for the total number of patent applications as well as for each of the three individual types of patent applications. Clearly, the positive relationship between FMA and firm innovation is not sensitive to how FMA is defined.

In summary, the robustness checks conducted in this section provide further evidence to support the hypothesis that FMA exert a positive impact on firm innovation. The concerns of endogeneity and definition of FMA have been addressed. It is reassuring to establish that the results of the robustness checks in this section corroborate the baseline results.

7. Conclusions

By merging China's manufacturing enterprise database, patent database and customs database, this paper constructs a sample of 27,130 firms

Table 8 Using 50% or More Foreign Ownership as FMA for Robustness Check

	<i>Total patents (p)</i>	<i>Invention (fm)</i>	<i>Utility model (sy)</i>	<i>Design (wg)</i>
	(1)	(2)	(3)	(4)
FMA 50%	0.432*	0.589*	0.456*	0.651***
Jb _{i,t-1}	(0.2601)	(0.3621)	(0.2754)	(0.1995)
TFP	0.224***	0.210***	0.152***	0.162**
Ln(tfp _{i,t-1})	(0.0540)	(0.0737)	(0.0570)	(0.0720)
K/L	0.075**	0.098**	0.010	-0.019
lnk+1	(0.0356)	(0.0423)	(0.0353)	(0.0489)
Subsidy	0.041	0.042	0.209***	0.082
Ln(sub+1)	(0.0729)	(0.0057)	(0.0701)	(0.0657)
Age	-0.090	-0.009	-0.039	-0.081
Ln(age+1)	(0.0618)	(0.1182)	(0.0930)	(0.0731)
Profit	-0.766**	-0.769	-0.380	-0.654
(pro)	(0.3876)	(0.5778)	(0.2940)	(0.4951)
Liability/	0.255	0.989***	0.229***	-0.652**
Asset	(0.2623)	(0.4212)	(0.2257)	(0.2750)
Firm FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Obs.	177,270	177,270	177,270	177,270

Notes: ***, ** and * indicate significance at the 1 per cent, 5 per cent and 10 per cent levels, respectively. Standard errors are in parentheses. The dependent variable is the total number of patent applications in column 1, invention patent applications in column 2, utility model patent applications in column 3 and design patent applications in column 4. Regressions use the Poisson fixed-effects panel estimator.

over the period 2002–2007. Using firm and year fixed-effects panel regressions to control for firm and year heterogeneities as well as adding a host of control variables, we find the following main results.

First, holding other variables constant, the average total number of patent applications of acquired firms is approximately 0.548 higher than that of domestic firms, and the average numbers of invention, utility model and design patent applications of acquired firms are approximately 0.796, 0.440 and 0.742 higher than those of domestic firms respectively. This provides clear evidence of a positive effect of FMA on firm innovation.

Second, FMA increase the patent applications of acquired firms through expanding firm size, exports and human capital formation. On average, a change in a firm's ownership from domestic to foreign in the previous year is estimated on average to associate with increases in firm size, exports and investment in on-the-job-training are 0.070 people, ¥2,740 and ¥200 respectively. Moreover, our results show that firm size, exports and human capital formation do not account for all of the effects of

FMA on firm innovation. There are additional effects of FMA on firm innovation that have not been accounted for by the mechanisms.

Finally, this article finds that wholly foreign-owned firms outperform partially foreign-owned firms in all types of patent applications. We also find that FMA have the largest positive effect on patent applications if the acquired firms are in labour-intensive industries. Technology-intensive firms, on the other hand, are the least affected by FMA in all patent applications.

The article's findings have obvious policy implications for China as well as for other developing countries at a similar stage of development. The findings suggest that it is possible to hasten the pace of technological upgrading of a latecomer country by welcoming FMA into the country. The relevant policies include reducing bureaucratic red tape on FMA, strengthening the protection of intellectual property rights and most of all ensuring the rule of law that provides effective protection of private property and a competitive environment for all firms.

June 2018

Appendix

Table A1 OLS Fixed-Effects Panel Estimations of FMA Effect on Firm Innovation

	<i>Total patents (lnp+1)</i>	<i>Invention (lnfin+1)</i>	<i>Utility model (lnsy+1)</i>	<i>Design (lnwg+1)</i>
	(1)	(2)	(3)	(4)
FMA	0.254***	0.078***	0.136***	0.077***
Jb _{i,t-1}	(0.0118)	(0.0064)	(0.0064)	(0.0092)
TFP	0.009***	0.004***	0.004***	0.004***
Ln(tfpi,t-1)	(0.0011)	(0.0006)	(0.0008)	(0.0007)
K/L	0.003**	0.001	0.001	0.001
lnk+1	(0.0012)	(0.0006)	(0.0007)	(0.0008)
Subsidy	0.003	0.001	0.002	0.001
Ln(sub+1)	(0.0019)	(0.0012)	(0.0012)	(0.0011)
Age	-0.005**	-0.002**	-1.57e-05	-0.004***
Ln(age+1)	(0.0020)	(0.0010)	(0.0012)	(0.0014)
Profit	-5.91e-05	2.65e-05	-2.42e-05	-4.12e-05
(pro)	(0.0001)	(0.00003)	(0.00004)	(0.0001)
Liability/	0.004	0.003**	0.001	0.001
Asset	(0.0033)	(0.0016)	(0.0021)	(0.0022)
Firm FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Obs.	162,780	162,780	162,780	162,780

Notes: ***, ** and * indicate significance at the 1 per cent, 5 per cent and 10 per cent levels, respectively. Heteroscedastic robust standard errors are in parentheses. The dependent variable is the total number of patent applications in column 1, invention patent applications in column 2, utility model patent applications in column 3 and design patent applications in column 4. Regressions use the OLS fixed-effects panel estimator.

Endnotes

1. Navaretti and Venables (2004) find that 89 per cent of FDI in developing countries was in the form of FMA compared to greenfield investment.
2. See, for example, Liu and Zou (2008, p. 352) who study Chinese high-tech industries to find that 'foreign greenfield R&D activities by multinational corporations in a host country significantly affect the innovation performance of domestic firms and there exist both intra-industry and inter-industry spill-overs from foreign greenfield R&D. There are only inter-industry M&A spill-overs'. Unlike our study, their study examines the spill-over effect of FMA on other firms rather than the effect of FMA on acquired firms.
3. A study by Chao and Yu (1996) finds that for a small open economy under tariff protection, the desirable policy is 100 per cent foreign ownership of subsidiaries, coupled with an export-share requirement.
4. The Poisson rather than the OLS fixed-effects estimator is used to estimate Equation (1) because the number of patent applications is a non-negative discrete random variable, and also a large number of firms have zero patent applications. The OLS assumption of the residual being normally distributed is thus violated. Following Cameron and Trivedi (2005), Hashmi (2013) and Aghion et al. (2013), we therefore adopt the Poisson fixed-effects

estimator. Results of the OLS fixed-effects estimation are also included in the Appendix for those who are interested in comparing the differences between the two estimators' estimates. In general, the OLS estimator's estimates are smaller in magnitude than those of the Poisson estimator, but they remain statistically significant.

5. Firms from industries 16, 23, 26, 27, 28, 29, 31, 34, 35, 36, 37, 38, 39 and 40 are technology-intensive; firms from industries 07, 15, 22, 25, 32, 44, 45 and 46 are capital-intensive; and firms from 06, 08, 09, 10, 13, 14, 17, 18, 19, 20, 21, 24, 30 and 33 are labour-intensive.

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